ENV 710 Initial Figures

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```r
# Load the packages
library(here)
library(tidyverse)
library(moments)
library(dplyr)
library(viridis)
library(ggplot2)
library(ggpath)
library(gt)
library(paletteer)
library(grid)
library(gridExtra)
library(scales)
library(kableExtra)
library(knitr)
library(ggcorrplot)
library(readr)
library(car)
library(lmerTest)
library(GGally)
library(sjPlot)
library(ggpubr)
library(performance)
library(see)
```

```r
# Set global options for R chunks
knitr::opts_chunk$set(echo = TRUE,
                      message = FALSE,
                      warning = FALSE)
```

Project files

The project is stored in the following Git repository: https://github.com/emilesflynn/ENV_710_Final_Project/

The raw data was downloaded from the Durham water quality web portal, which is managed by the City of Durham: http://www.durhamwaterquality.org/

Data Wrangling

The raw data frame presents each measurement in a separate row. The code below merge the measurements conducted at the same location on the same day into one observation.

```
# Set ggplot theme
mytheme <- theme_classic(base_size = 14) +
  theme(axis.text = element_text(color = "black"),
        legend.position = "right")
theme_set(mytheme)

# Load datasets
durham_data_raw <- read_csv(here("Data/SlaySMilesFlynnESuEData_Raw_2025.csv"),
    col_types = cols(Date = col_date(format = "%Y/%m/%d")))
# View(durham_data_raw)

# Pivot table and merge the measurements
df_wide <- durham_data_raw %>%
  pivot_wider(
    id_cols = c(`Station Name`, Date),
    names_from = Parameter,
    values_from = Value,
    values_fn = mean
  )

write.csv(df_wide, file = here("Data/SlaySMilesFlynnESuEData_Cleaned.csv"), row.names = FALSE)
```

After obtaining the clean data frame, we then went on to add more explanatory or informational columns to help with further analyses. We isolated the river name and the sampling locations' distances to the river mouth, and add the year and month columns to help visualize the trends on the temporal scale.

```
# Load the datasets
here()
```

```
## [1] "/Users/ericarquette/ENV710Project"
```

```
data_clean <- read_csv(here("Initial Figures/SlaySMilesFlynnESuEData_Cleaned_2025.csv"))

data_clean <- data_clean %>%
  mutate(
    #Extract the river name abbreviation from the station name
    River_abbr = str_extract(`Station Name`, "^[A-Z]{2}"),
    #Extract the distance to stream mouth
    Distance_miles = as.numeric(str_extract(`Station Name`, "[0-9.]+"))
  )

# Find rows where the distance extraction failed
problem_rows <- data_clean%>%
  mutate(temp_dist = str_extract(`Station Name`, "[0-9.]+")) %>%
  filter(is.na(as.numeric(temp_dist)) & !is.na(`Station Name`))

# Show the unique station names that failed
unique(problem_rows$`Station Name`)
```

```
## character(0)
```

```r
# Replace the abbreviations with actual names
river_lookup <- data.frame(
  River_abbr = c("EL", "EN", "NH", "TF", "CC", "NE", "LC", "LL", "LR", "PN", "SI"),
  Watershed = c("Ellerbe Creek", "Eno River", "New Hope Creek", "Third Fork Creek",
                      "Crooked Creek", "Northeast Creek", "Lick Creek", "Little Lick Creek",
                      "Little River", "Panther Creek", "Stirrup Iron Creek")
)

# Merge the data frames
data_clean_final <- data_clean %>%
  left_join(river_lookup, by = "River_abbr")

# Add month and year columns
data_clean_final$Date <- ymd(data_clean_final$Date)

data_clean_final <- data_clean_final %>%
  mutate(Month = factor(month(Date),
                        levels = 1:12,
                        labels = month.abb,
                        ordered = TRUE),
         Year = year(Date),
         Storm_status = case_when(Date <= as.Date("2025-07-03") ~ "Before",
                                  Date >= as.Date("2025-07-09") ~ "After"),
         Basin = case_when(
           River_abbr %in% c("EL", "EN", "LC", "LL", "LR", "PN", "SI") ~ "Neuse (North)",
           River_abbr %in% c("CC", "NE", "NH",  "TF") ~ "Cape Fear (South)"))

# Reorder columns
data_clean_final_2025 <- data_clean_final %>%
  select(`Station Name`, Watershed, River_abbr, Distance_miles, Basin,
         Date, Month, Year, Storm_status, everything())

write.csv(data_clean_final_2025, file = here("Data/SlaySMilesFlynnESuEData_Final_2025.csv"),
          row.names = FALSE)

# Combined the dataset
# Change the year above to 2024 to obtain the 2024 data frame
data_2024 <- read.csv(here("Data/SlaySMilesFlynnESuEData_Final_2024.csv"))
data_2025 <- read.csv(here("Data/SlaySMilesFlynnESuEData_Final_2025.csv"))
data_combined_2425 <- bind_rows(data_2024, data_2025) %>%
  mutate(Date = as.Date(Date),
         Year = as.factor(year(Date)),
         # Extract the numeric month (1-12) for the x-axis calculation
         Month_Num = month(Date))
```

Intial figures

Trends over the time

This figure shows how the concentration of Dissolved Oxygen (mg/L) changed over time before and after the storm. The vertical line indicates when the Tropical Storm Chantal occurred on July 6, 2025.

```r
# Comparing between Northern and Southern Basins
do_plot_2025 <- ggplot(data_clean_final_2025, aes(x = Date, y = `Dissolved Oxygen`,
                                                  color = Basin)) +
  # Add points and lines (grouped by Basin to show trends)
  geom_point(alpha = 0.5, na.rm = TRUE) +
  geom_smooth(se = FALSE, method = "loess", na.rm = TRUE) +

  # Add the NC DEQ Water Quality Standard line (5.0 mg/L)
  geom_hline(yintercept = 5.0, linetype = "dashed", color = "darkorange") +
  annotate("text", x = min(data_clean_final$Date), y = 5.5,
           label = "NC Standard (5.0 mg/L)", color = "red", hjust = 0) +

  # Add a vertical line to represent TS Chantal
  geom_vline(xintercept = as.Date("2025-07-06"),
             linetype = "dashed",
             color = "darkred",
             linewidth = 0.5) +
  # Add a text label next to the line
  annotate("text",
           x = as.Date("2025-07-06"),
           y = 11,
           label = "TS Chantal (Jul 6)",
           color = "darkred",
           angle = 90,
           vjust = -0.5,
           fontface = "bold")+

  # Extend the x-axis to show the full labels
  scale_x_date(
    limits = c(as.Date("2025-01-01"), as.Date("2026-01-01")),
    date_labels = "%b %Y",
    date_breaks = "4 months")+

  # Formatting
  scale_color_manual(values = c("Neuse (North)" = "#2c7bb6",
                                "Cape Fear (South)" = "#d7191c")) +
  labs(title = "Dissolved Oxygen Levels in Durham Streams (2025)",
       subtitle = "Comparing Northern (Neuse) and Southern (Cape Fear) Watersheds",
       x = "Sampling Date",
       y = "Dissolved Oxygen (mg/L)") +
  theme_minimal() +
  theme(legend.position = "bottom")
do_plot_2025
```

Scatterplot for linear regression models

The figure below shows the relationship between Total Phosphorus (TP), which is a variable showing aquatic nutrient pollution level, and Dissolved Oxygen (DO), which is a strong aquatic health indicator. The linear regression model finds a strong relationship (*p-value* < 0.05).

Note: Why are we studying how to predict DO? Why don't we directly measure DO?

Answer: DO is a good indicator for water health; however, since the resulting DO can be attributed to many factors, the DO level may result from different environments and similar environments might lead to

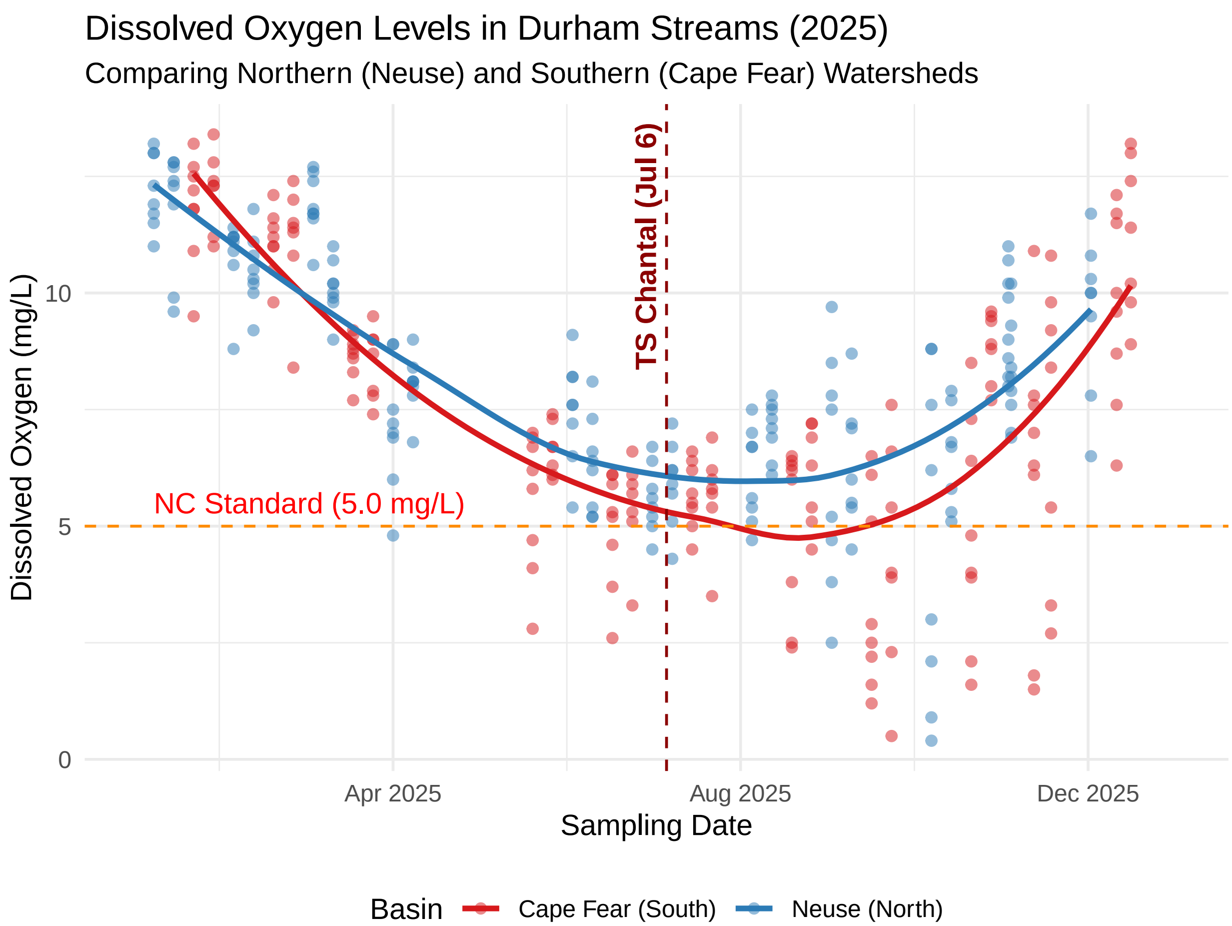


Figure 1: 2025 Durham Stream Dissolved Oxygen Levels Streams by Basin.

different DO levels. Testing Dissolved Oxygen (DO) directly only tells you the "what," while this analysis is designed to tell you the "why."

By studying how the environmental factors/parameters contribute to the DO level, we can better understand:

(1) How sensitive is DO to the environmental changes? How does DO change?

(2) How might the DO level/water quality change in the future?

```r
regression_plot <- data_clean_final_2025  %>%
  # This removes all rows where the x-variable is greater than 0.65
  filter(`Total Phosphorus` <= 0.65) %>%
  ggplot(aes(x = `Total Phosphorus`, y = `Dissolved Oxygen`,
                                          color = Basin)) +
  # Use points with transparency to handle overlapping data
  geom_point(alpha = 0.5, size = 2, na.rm = TRUE) +

  # Add the Linear Regression Model (method = "lm")
  # This adds a line of best fit for each Basin
  geom_smooth(method = "lm", se = TRUE, fill = "lightgray", na.rm = TRUE) +

  # Formatting
  scale_color_brewer(palette = "Set1") +
  # facet_wrap(~Basin) +
  labs(title = "Relationship Between Total Phosphorus and Dissolved Oxygen",
       subtitle = "Linear regression of Total Phosphorus vs. Dissolved Oxygen by Basin",
       x = "Total Phosphorus (mg/L)",
       y = "Dissolved Oxygen (mg/L)",
       color = "Basin",
       caption = "Three outliers (TP > 0.65 mg/L) removed as high-leverage observations") +
  theme_minimal() +
  theme(legend.position = c(0.8, 0.15))

regression_plot
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: 'Dissolved Oxygen' ~ Basin + Storm_Period + Distance_miles +
##     (1 | River_Name/Station.Name)
##    Data: combined_data
##
## REML criterion at convergence: 1563
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -2.39859 -0.65925 -0.06632  0.77380  2.43163
##
## Random effects:
##  Groups                 Name        Variance Std.Dev.
##  Station.Name:River_Name (Intercept) 0.2264   0.4759
##  River_Name             (Intercept) 0.4964   0.7046
##  Residual                           6.1844   2.4868
## Number of obs: 331, groups:  Station.Name:River_Name, 31; River_Name, 11
```

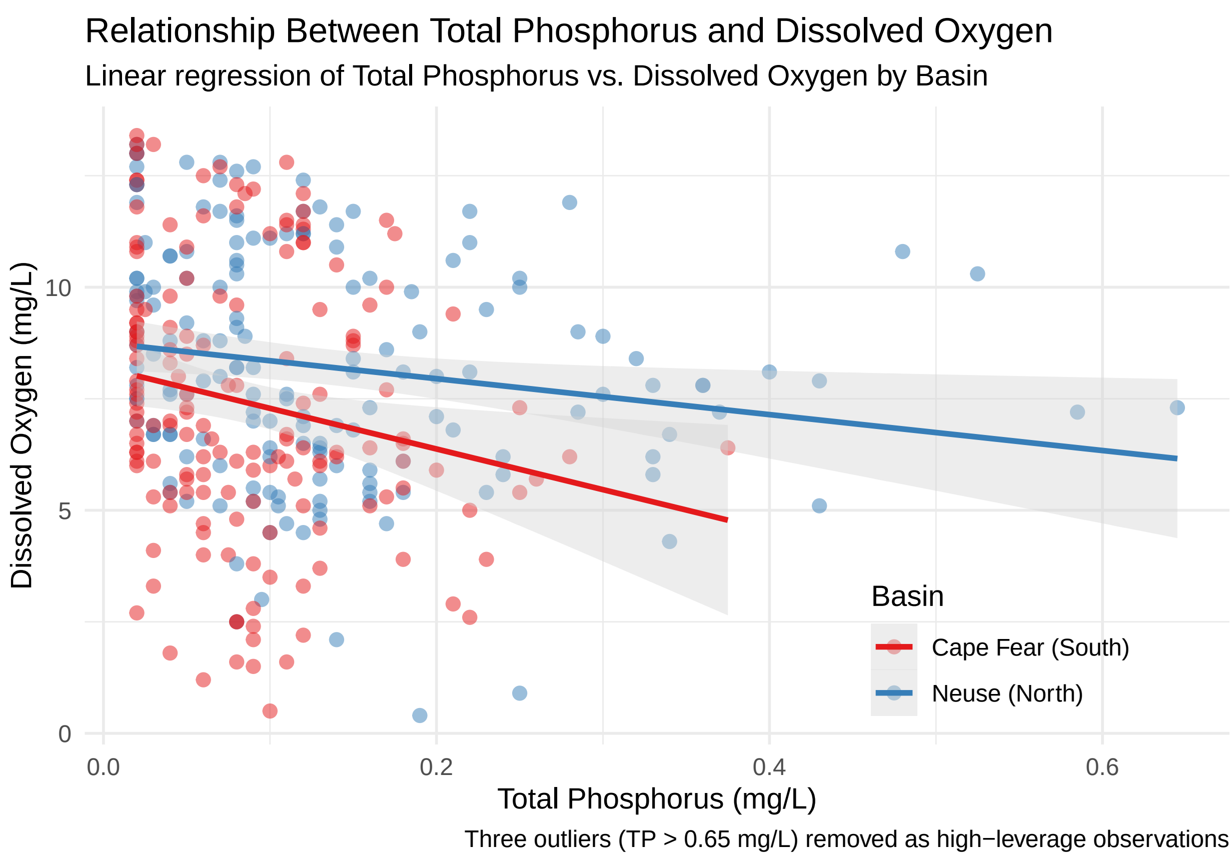


Figure 2: Linear regression of Total Phosphorus v. Dissolved Oxygen by Basin.

```
##
## Fixed effects:
##                          Estimate Std. Error         df t value Pr(>|t|)
## (Intercept)               8.56677    0.49723    9.08276  17.229 3.02e-08 ***
## BasinNeuse (North)        0.98021    0.60117    8.26196   1.630    0.140
## Storm_PeriodPost-Storm   -2.43047    0.27416  299.29357  -8.865  < 2e-16 ***
## Distance_miles           -0.01085    0.07082   27.48420  -0.153    0.879
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##             (Intr) BsN(N) S_PP-S
## BsnNs(Nrth) -0.533
## Strm_PrdP-S -0.302  0.014
## Distanc_mls -0.366 -0.308  0.002
```

```
## Type III Analysis of Variance Table with Satterthwaite's method
##                Sum Sq Mean Sq NumDF   DenDF F value Pr(>F)
## Basin           16.44   16.44     1   8.262  2.6585 0.1404
## Storm_Period   486.04  486.04     1 299.294 78.5914 <2e-16 ***
## Distance_miles   0.15    0.15     1  27.484  0.0235 0.8794
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Analyses

Part 1: Mann-Whitney U Test of DO between N and S basins.

Note p-value of Mann-Whitney U test is 0.01 < 0.05, which suggests a significant difference.

What does it mean/what are the implications? -> the two watersheds do not respond to environmental changes in the same way, and thus we/the city should design specific strategies for each of them.

```r
test_data <- combined_data %>%
  filter(!is.na(`Dissolved Oxygen`))

# 2. Check for Normality (Shapiro-Wilk Test)
# If p < 0.05, the data is NOT normal
shapiro_north <- shapiro.test(test_data$`Dissolved Oxygen`[test_data$Basin == "Neuse (North)"])
shapiro_south <- shapiro.test(test_data$`Dissolved Oxygen`[test_data$Basin == "Cape Fear (South)"])

print(paste("Shapiro P-value (North):", shapiro_north$p.value))
```

```
## [1] "Shapiro P-value (North): 0.0165897835077109"
```

```r
print(paste("Shapiro P-value (South):", shapiro_south$p.value))
```

```
## [1] "Shapiro P-value (South): 0.00525282619798975"
```

```r
# 3. Check for Homogeneity of Variance (Levene's Test)
# If p < 0.05, variances are significantly different
levene_result <- leveneTest(`Dissolved Oxygen` ~ Basin, data = test_data)
print(levene_result)
```

```
## Levene's Test for Homogeneity of Variance (center = median)
##        Df F value  Pr(>F)
## group   1  4.7852 0.02941 *
##       329
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
# Logic: Use t-test if both Shapiro p > 0.05 AND Levene p > 0.05. Otherwise, use Wilcoxon.
if(shapiro_north$p.value > 0.05 & shapiro_south$p.value > 0.05 & levene_result$`Pr(>F)`[1] > 0.05) {
  print("Assumptions met: Performing Welch's T-test")
  final_test <- t.test(`Dissolved Oxygen` ~ Basin, data = test_data)
} else {
  print("Assumptions NOT met: Performing Mann-Whitney U Test (Wilcoxon Rank Sum)")
  final_test <- wilcox.test(`Dissolved Oxygen` ~ Basin, data = test_data)
}
```

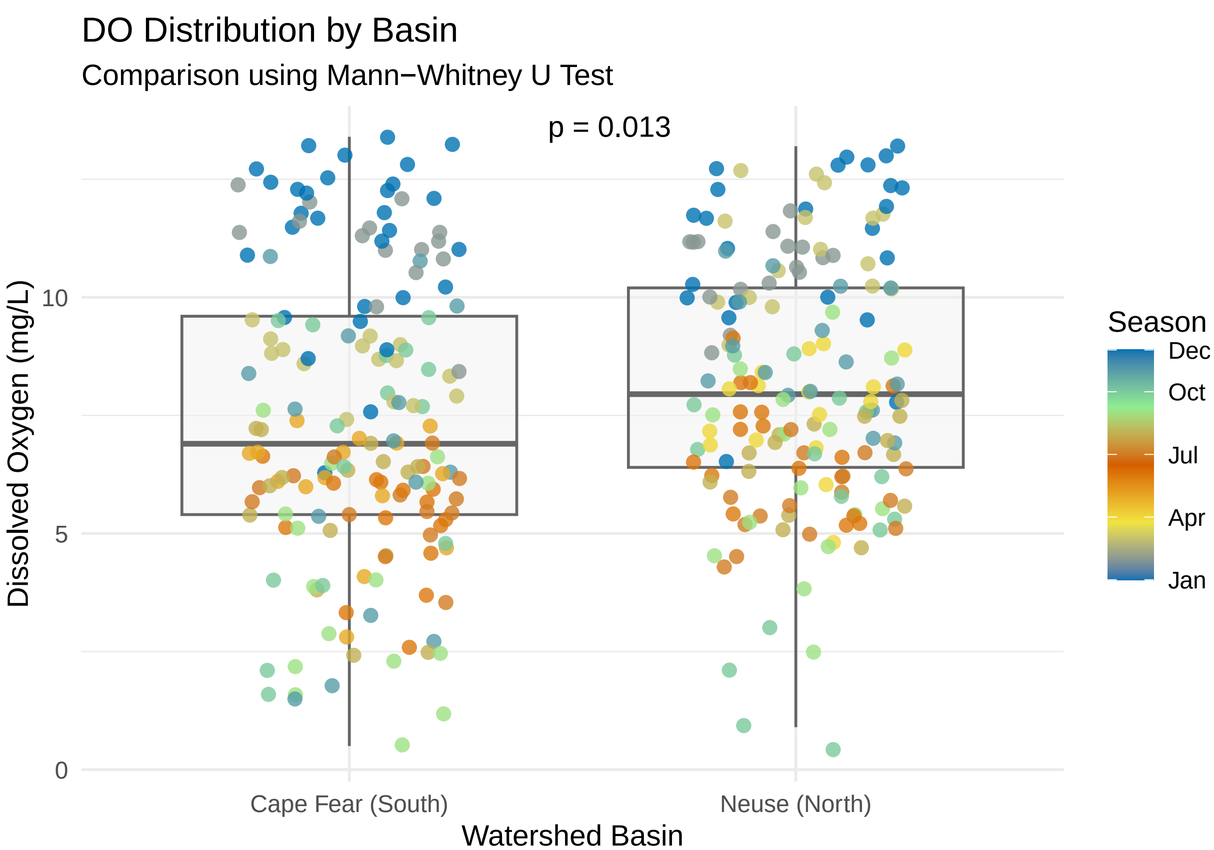
```
## [1] "Assumptions NOT met: Performing Mann-Whitney U Test (Wilcoxon Rank Sum)"
```

```
# Filter by treatment type
test_N <- test_data %>%
filter(Basin== "Neuse (North)")
test_S <- test_data %>%
filter(Basin== "Cape Fear (South)")
final_test1 <- wilcox.test(test_N$`Dissolved Oxygen`, test_S$`Dissolved Oxygen`,
                           paired = FALSE)

print(final_test1)
```

```
##
##  Wilcoxon rank sum test with continuity correction
##
## data:  test_N$`Dissolved Oxygen` and test_S$`Dissolved Oxygen`
## W = 15846, p-value = 0.01349
## alternative hypothesis: true location shift is not equal to 0
```

```
model_data1 <- combined_data %>%
  mutate(Month_Num = as.numeric(factor(Month,
                                levels = c("Jan", "Feb", "Mar", "Apr", "May", "Jun",
                                           "Jul", "Aug", "Sep", "Oct", "Nov", "Dec"))))

ggplot(model_data1, aes(x = Basin, y = `Dissolved Oxygen`)) +
  # Boxplots (Keeping them neutral to let the points shine)
  geom_boxplot(fill = "grey95", color = "grey40", outlier.shape = NA, alpha = 0.5) +

  # Jittered points colored by Month (using the numeric gradient)
  geom_jitter(aes(color = Month_Num), width = 0.25, alpha = 0.8, size = 2) +

  # The Red-Blue Seasonal Scale
  # Low (Jan) = Blue, Mid (July) = Red, High (Dec) = Blue
  # We use a n-color gradient to create a 'loop' or use a mid-point
  scale_color_gradientn(colors = c("#0072B2", "#F0E442", "#D55E00", "#90EE90", "#0072B2"),
                        breaks = c(1, 4, 7, 10, 12),
```

```r
                          labels = c("Jan", "Apr", "Jul", "Oct", "Dec"),
                          name = "Season") +

  # Statistics
  stat_compare_means(method = "wilcox.test", label = "p.format",
                       label.x = 1.5) +

  theme_minimal() +
  labs(title = "DO Distribution by Basin",
       subtitle = "Comparison using Mann-Whitney U Test",
       x = "Watershed Basin",
       y = "Dissolved Oxygen (mg/L)")
```



Part 2. Mixed models

First we need to check the correlation to make avoid multicollinearity. Use the criterion mentioned in lab, correlation > 0.6. And later, after we built the model, we can double check the variance using: Variance Inflation Factor (VIF), to be mentioned in the Discussions.

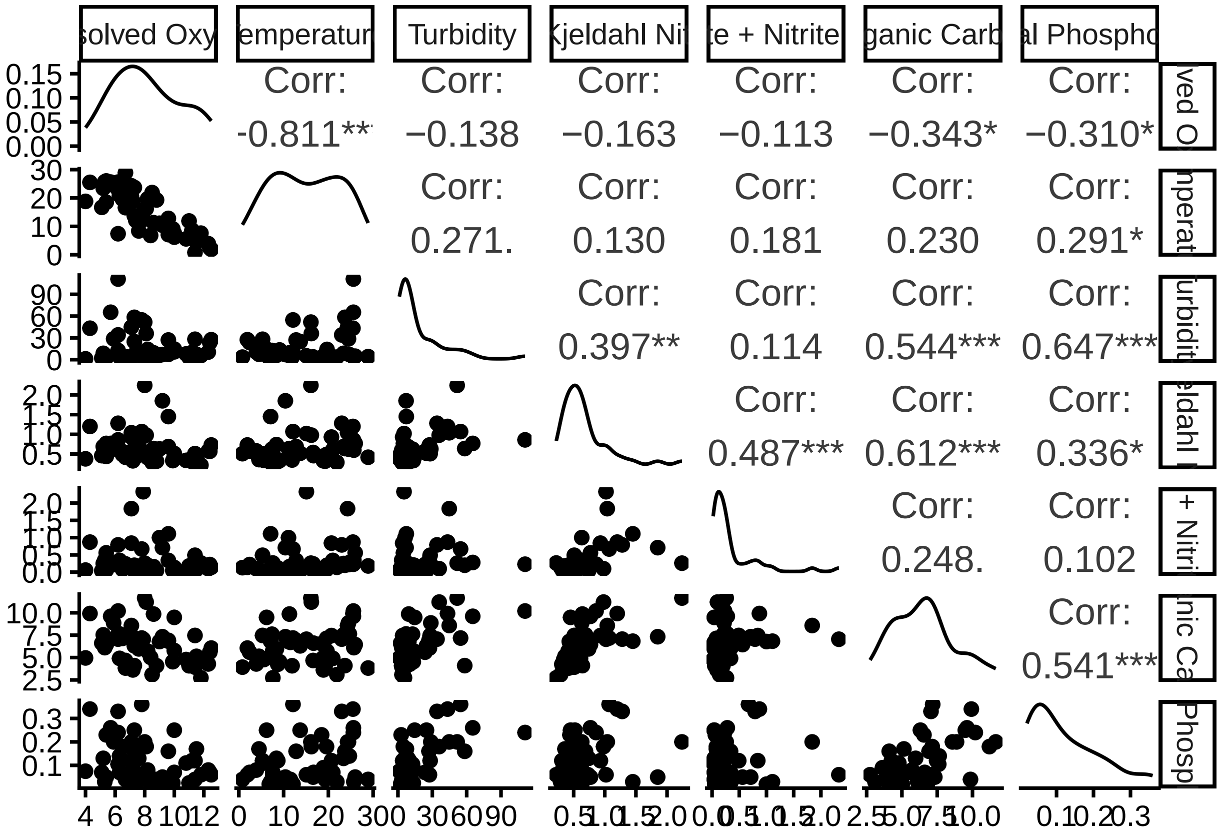
```r
# 1. Check for correlation first
cor_check <- combined_data %>%
  select(`Dissolved Oxygen`, Temperature, Turbidity) %>%
  drop_na() %>%
  cor()
print(cor_check)
```

```
##                   Dissolved Oxygen Temperature  Turbidity
## Dissolved Oxygen       1.00000000  -0.7123896 0.01973114
## Temperature           -0.71238963   1.0000000 0.11207354
## Turbidity              0.01973114   0.1120735 1.00000000
```

```r
# 1. Select candidate parameters
params <- combined_data %>%
  select(`Dissolved Oxygen`, Temperature, Turbidity,
         `Total Kjeldahl Nitrogen`, `Nitrate + Nitrite as N`, `Organic Carbon`,
         `Total Phosphorus`) %>%
  drop_na()

# 2. Generate the correlation matrix
cor_matrix <- cor(params)
print(cor_matrix)
```

```
##                          Dissolved Oxygen Temperature  Turbidity
## Dissolved Oxygen                1.0000000  -0.8113451 -0.1384645
## Temperature                    -0.8113451   1.0000000  0.2706821
## Turbidity                      -0.1384645   0.2706821  1.0000000
## Total Kjeldahl Nitrogen        -0.1630389   0.1303473  0.3969996
## Nitrate + Nitrite as N         -0.1132157   0.1809631  0.1144985
## Organic Carbon                 -0.3432377   0.2299837  0.5442105
## Total Phosphorus               -0.3097656   0.2906731  0.6465871
##                          Total Kjeldahl Nitrogen Nitrate + Nitrite as N
## Dissolved Oxygen                      -0.1630389             -0.1132157
## Temperature                            0.1303473              0.1809631
## Turbidity                              0.3969996              0.1144985
## Total Kjeldahl Nitrogen                1.0000000              0.4874678
## Nitrate + Nitrite as N                 0.4874678              1.0000000
## Organic Carbon                         0.6117778              0.2475691
## Total Phosphorus                       0.3364888              0.1020310
##                          Organic Carbon Total Phosphorus
## Dissolved Oxygen             -0.3432377       -0.3097656
## Temperature                   0.2299837        0.2906731
## Turbidity                     0.5442105        0.6465871
## Total Kjeldahl Nitrogen       0.6117778        0.3364888
## Nitrate + Nitrite as N        0.2475691        0.1020310
## Organic Carbon                1.0000000        0.5410604
## Total Phosphorus              0.5410604        1.0000000
```

```r
# 3. Visual Check
ggpairs(params)
```



Then, we compare the different model candidates. We compare AIC to decide which model is better.

```r
# This ensures all models see the exact same rows
model_subset <- combined_data %>%
  drop_na(`Dissolved Oxygen`, Temperature, `Organic Carbon`,
          `Total Phosphorus`, Turbidity, `Total Kjeldahl Nitrogen`)
# Hypothesis: DO is driven by temperature and organic nutrient loading.
model_1 <- lmer(`Dissolved Oxygen` ~ Basin + Temperature + `Organic Carbon` + `Total Phosphorus` +
                (1 | River_Name), data = model_subset, REML = FALSE)
# Hypothesis: DO is driven by temperature, physical sediment, and nitrogenous waste.
model_2 <- lmer(`Dissolved Oxygen` ~ Basin + Temperature + Turbidity + `Total Kjeldahl Nitrogen` +
                (1 | River_Name), data = model_subset, REML = FALSE)
# Hypothesis: Temperature and Organic Carbon are the primary drivers.
model_3 <- lmer(`Dissolved Oxygen` ~ Basin + Temperature + `Organic Carbon` +
                (1 | River_Name), data = model_subset, REML = FALSE)
# Compare AIC (Lower is better)
anova(model_1, model_2, model_3)
```

```
## Data: model_subset
## Models:
## model_3: 'Dissolved Oxygen' ~ Basin + Temperature + 'Organic Carbon' + (1 | River_Name)
## model_1: 'Dissolved Oxygen' ~ Basin + Temperature + 'Organic Carbon' + 'Total Phosphorus' + (1 | Rive
## model_2: 'Dissolved Oxygen' ~ Basin + Temperature + Turbidity + 'Total Kjeldahl Nitrogen' + (1 | Rive
##         npar    AIC    BIC  logLik deviance  Chisq Df Pr(>Chisq)
## model_3    6 175.72 187.31 -81.858   163.72
## model_1    7 177.71 191.24 -81.857   163.71 0.0022  1     0.9623
## model_2    7 174.34 187.87 -80.171   160.34 3.3714  0
```

Now, the model I am showing below is DO~ Basin (categorical) + Temp + Turbidity + TKN + Stream (partial pooling)

```r
# Final Production Model
final_model <- lmer(`Dissolved Oxygen` ~ Basin + Temperature + Turbidity + `Total Kjeldahl Nitrogen` +
                     (1 | River_Name), data = combined_data, REML = TRUE)

summary(final_model)
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula:
## `Dissolved Oxygen` ~ Basin + Temperature + Turbidity + `Total Kjeldahl Nitrogen` +
##     (1 | River_Name)
##    Data: combined_data
##
## REML criterion at convergence: 1300.3
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -3.5267 -0.4593  0.2330  0.6635  2.1251
##
## Random effects:
##  Groups     Name        Variance Std.Dev.
##  River_Name (Intercept) 0.7877   0.8875
##  Residual               3.2763   1.8101
## Number of obs: 315, groups:  River_Name, 11
##
## Fixed effects:
##                            Estimate Std. Error         df t value Pr(>|t|)
## (Intercept)               11.009710   0.537063  11.674724  20.500 1.62e-10 ***
## BasinNeuse (North)         1.309305   0.609771   7.872053   2.147  0.06460 .
## Temperature               -0.270669   0.014025 304.239491 -19.300  < 2e-16 ***
## Turbidity                  0.007176   0.002283 309.386442   3.143  0.00184 **
## `Total Kjeldahl Nitrogen`  -0.522904   0.378515 309.908818  -1.381  0.16813
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##             (Intr) BsN(N) Tmprtr Trbdty
## BsnNs(Nrth) -0.682
## Temperature -0.251 -0.028
## Turbidity   -0.001 -0.061 -0.151
## `TtlKNtrgn` -0.315  0.017 -0.233 -0.055
```

```r
coef_plot <- plot_model(final_model,
                        type = "est",
                        show.values = TRUE,
                        value.offset = .3,
                        title = "Predictors of Dissolved Oxygen in Durham Streams",
                        axis.labels = c("TKN (Nutrients)", "Turbidity (Runoff)",
                                        "Temperature", "Basin [Neuse]"),
                        vline.color = "grey")
```

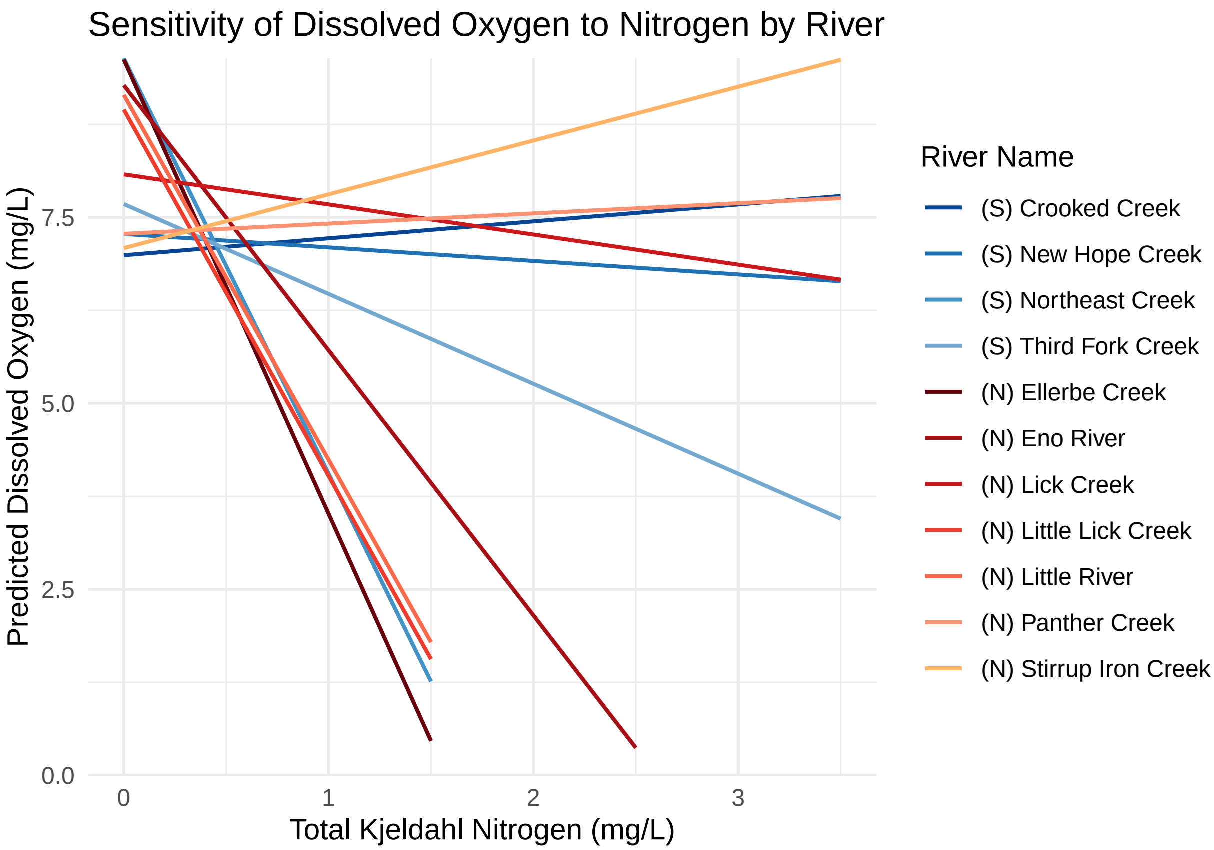
```r
effect_plot <- plot_model(final_model,
                          type = "pred",
                          terms = c("Total Kjeldahl Nitrogen", "Basin"),
                          title = "Predicted Dissolved Oxygen Levels",
                          axis.title = c("Water Temperature (°C)", "Total Kjeldahl Nitrogen"))
```

Note (1) difference between N and S basins, (2) the variables that are significant in the model.

```r
# We allow TKN to have a random slope: (Total.Kjeldahl.Nitrogen | River_Name)
slope_model_tn <- lmer(`Dissolved Oxygen` ~ Basin + Temperature + Turbidity+
                          `Total Kjeldahl Nitrogen` +
                        (`Total Kjeldahl Nitrogen` | River_Name),
                        data = combined_data, control = lmerControl(optimizer = "bobyqa"))

summary(slope_model_tn)
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula:
## `Dissolved Oxygen` ~ Basin + Temperature + Turbidity + `Total Kjeldahl Nitrogen` +
##     (`Total Kjeldahl Nitrogen` | River_Name)
##    Data: combined_data
## Control: lmerControl(optimizer = "bobyqa")
##
## REML criterion at convergence: 1279.4
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -3.6207 -0.4835  0.2152  0.6711  2.3294
##
## Random effects:
##  Groups     Name                      Variance Std.Dev. Corr
##  River_Name (Intercept)               1.725    1.313
##             `Total Kjeldahl Nitrogen` 9.561    3.092    -0.94
##  Residual                             2.979    1.726
## Number of obs: 315, groups:  River_Name, 11
##
## Fixed effects:
##                            Estimate Std. Error         df t value Pr(>|t|)
## (Intercept)               11.862551   0.548644   8.176789  21.622 1.66e-08 ***
## BasinNeuse (North)         1.163623   0.386439   6.263955   3.011  0.02245 *
## Temperature               -0.260049   0.013741 304.024742 -18.925  < 2e-16 ***
## Turbidity                  0.007072   0.002157 286.904878   3.278  0.00117 **
## `Total Kjeldahl Nitrogen` -2.344395   1.096489   8.007624  -2.138  0.06494 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##             (Intr) BsN(N) Tmprtr Trbdty
## BsnNs(Nrth) -0.366
## Temperature -0.192 -0.032
## Turbidity    0.012 -0.078 -0.127
## `TtlKNtrgn` -0.792 -0.013 -0.137 -0.043
```

```r
# This ensures they are sorted by Basin in the legend
combined_data_sorted <- combined_data %>%
  mutate(Basin_Code = ifelse(Basin == "Neuse (North)", "N", "S")) %>%
  mutate(Legend_Name = paste0("(", Basin_Code, ") ", River_Name)) %>%
  arrange(Basin, River_Name)

# 2. Extract the sorted levels for the legend
sorted_labels <- unique(combined_data_sorted$Legend_Name)

# 3. Define the 11-color palette (7 Reds for North, 4 Blues for South)
custom_colors <- c(
  "#084594", "#2171b5", "#4292c6", "#74a9cf",                          # Cape Fear
  "#67000d", "#a50f15", "#cb181d", "#ef3b2c", "#fb6a4a", "#fc9272", "#ffb366" # Neuse
)

# This will plot the individual slopes for each river
tn_slope_plot <- plot_model(slope_model_tn,
                            type = "pred",
                            terms = c("Total Kjeldahl Nitrogen", "River_Name"),
                            pred.type = "re", # This ensures it uses the random effects (various slopes
                            ci.lvl = NA) +    # Removing CI ribbons makes the slopes easier to see
  scale_color_manual(values = custom_colors,
                     labels = sorted_labels) +
  # Force the Y-axis to start at 0 but let the top be automatic
  scale_y_continuous(limits = c(0, NA), expand = c(0, 0)) +
  labs(title = "Sensitivity of Dissolved Oxygen to Nitrogen by River",
       x = "Total Kjeldahl Nitrogen (mg/L)",
       y = "Predicted Dissolved Oxygen (mg/L)",
       colour = "River Name") +
  #scale_color_viridis_d(option = "plasma") +
  theme_minimal()

print(tn_slope_plot)
```




```r
# 4. Check the "Pooling" effect (Random Effects variance)
# This shows how much variance is explained by the differences between rivers
print(VarCorr(slope_model_tn), comp = "Variance")
```

```
##  Groups     Name                       Variance Cov
##  River_Name (Intercept)                1.7249
##             `Total Kjeldahl Nitrogen` 9.5607   -3.815
##  Residual                              2.9788
```

Note: How each river is responding to the environmental changes differently.

Our model revealed significant variation in how Durham's rivers respond to nitrogen stress. With a TKN variance of 9.56, we see that river sensitivity is not uniform; some streams are far more vulnerable to nutrient pulses than others. Interestingly, the negative covariance suggests that our highest-quality baseline sites actually show the steepest declines under stress, highlighting a critical need to protect our remaining high-functioning riparian zones.

Below we are adding the storm event into the model:

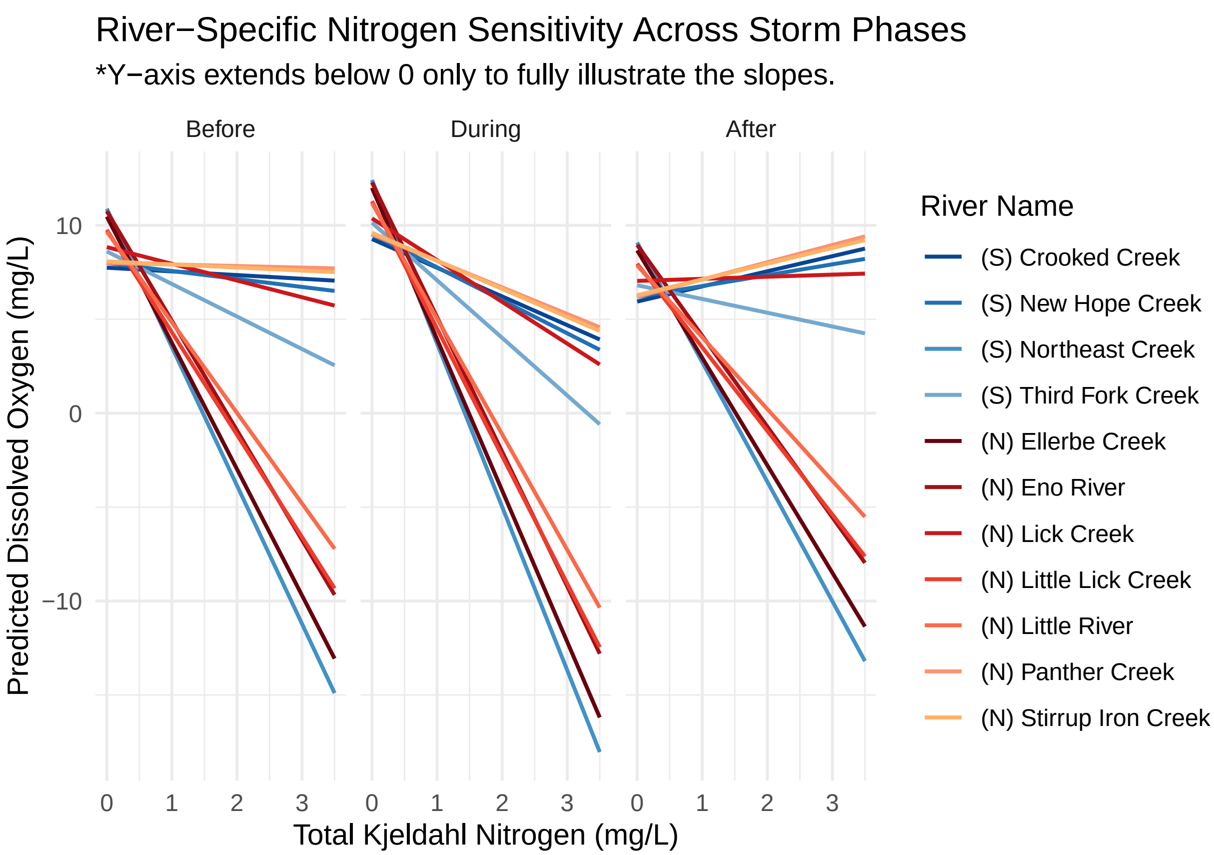
```r
# Define the storm windows
combined_data <- combined_data %>%
  mutate(Date = as.Date(Date),
         Storm_Phase = case_when(
           Date < as.Date("2025-07-01") ~ "Before",
           Date >= as.Date("2025-07-01") & Date <= as.Date("2025-07-31") ~ "During",
           Date > as.Date("2025-07-31") ~ "After"
         ),
```

```r
        # Set "Before" as the reference group for comparison
        Storm_Phase = factor(Storm_Phase, levels = c("Before", "During", "After")))

# We use the fixed effects from "Winner" Model 2
# We add Storm_Phase and its interaction with TKN
storm_impact_model <- lmer(`Dissolved Oxygen` ~ Basin + Temperature + Turbidity +
                           `Total Kjeldahl Nitrogen` * Storm_Phase +
                           (`Total Kjeldahl Nitrogen`| River_Name),
                           data = combined_data)

summary(storm_impact_model)
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula:
## `Dissolved Oxygen` ~ Basin + Temperature + Turbidity + `Total Kjeldahl Nitrogen` *
##     Storm_Phase + (`Total Kjeldahl Nitrogen` | River_Name)
##    Data: combined_data
##
## REML criterion at convergence: 1224
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -3.4690 -0.4953  0.0741  0.6124  2.5002
##
## Random effects:
##  Groups     Name                      Variance Std.Dev. Corr
##  River_Name (Intercept)                1.98    1.407
##             `Total Kjeldahl Nitrogen` 11.09    3.331    -0.95
##  Residual                              2.52    1.587
## Number of obs: 315, groups:  River_Name, 11
##
## Fixed effects:
##                                              Estimate Std. Error         df
## (Intercept)                                 12.855803   0.595754  11.315144
## BasinNeuse (North)                           1.054784   0.366760   6.271071
## Temperature                                 -0.264448   0.014275 296.514716
## Turbidity                                    0.005413   0.002014 282.438111
## `Total Kjeldahl Nitrogen`                   -3.062766   1.198729   9.572968
## Storm_PhaseDuring                            1.527020   1.272966 298.252820
## Storm_PhaseAfter                            -1.800534   0.437257 298.347603
## `Total Kjeldahl Nitrogen`:Storm_PhaseDuring  -1.331925   1.709239 298.581931
## `Total Kjeldahl Nitrogen`:Storm_PhaseAfter    1.001086   0.699744 298.611015
##                                             t value Pr(>|t|)
## (Intercept)                                  21.579 1.50e-10 ***
## BasinNeuse (North)                           2.876  0.02685 *
## Temperature                                 -18.526  < 2e-16 ***
## Turbidity                                    2.687  0.00763 **
## `Total Kjeldahl Nitrogen`                   -2.555  0.02954 *
## Storm_PhaseDuring                            1.200  0.23126
## Storm_PhaseAfter                            -4.118 4.96e-05 ***
## `Total Kjeldahl Nitrogen`:Storm_PhaseDuring  -0.779  0.43645
## `Total Kjeldahl Nitrogen`:Storm_PhaseAfter    1.431  0.15358
```

```
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##              (Intr) BsN(N) Tmprtr Trbdty `TKNt` Str_PD Str_PA `TKN`:S_PD
## BsnNs(Nrth) -0.339
## Temperature -0.156 -0.042
## Turbidity   -0.006 -0.075 -0.056
## `TtlKNtrgn` -0.833  0.003 -0.105 -0.038
## Strm_PhsDrn -0.051 -0.034 -0.207 -0.084  0.089
## Strm_PhsAft -0.336  0.075 -0.138  0.000  0.260  0.162
## `TKNt`:S_PD  0.094  0.043  0.080  0.052 -0.119 -0.955 -0.160
## `TKNt`:S_PA  0.320 -0.066  0.061  0.023 -0.290 -0.129 -0.896  0.175
## optimizer (nloptwrap) convergence code: 0 (OK)
## Model failed to converge with max|grad| = 0.00412688 (tol = 0.002, component 1)
```

```r
# We use type = "pred" to show predicted values
# We facet by Storm_Phase to see how the plot changes over time
spaghetti_storm <- plot_model(storm_impact_model,
                              type = "pred",
                              terms = c("Total Kjeldahl Nitrogen", "River_Name", "Storm_Phase"),
                              pred.type = "re", # Use random effects for individual river lines
                              ci.lvl = NA) +    # Remove ribbons for clarity
  #scale_color_viridis_d(option = "turbo") +
  scale_color_manual(values = custom_colors,
                     labels = sorted_labels) +
  # Force the Y-axis to start at 0 but let the top be automatic
  #scale_y_continuous(limits = c(0, NA), expand = c(0, 0)) +
  labs(title = "River-Specific Nitrogen Sensitivity Across Storm Phases",
       subtitle = "*Y-axis extends below 0 only to fully illustrate the slopes.",
       x = "Total Kjeldahl Nitrogen (mg/L)",
       y = "Predicted Dissolved Oxygen (mg/L)",
       colour = "River Name") +
  theme_minimal() +
  facet_wrap(~facet) # This creates three panels: Before, During, and After

print(spaghetti_storm)
```



What we can learn from the new model with storm:

Key Interpretations of Fixed Effects

The Temperature Law: As expected, Temperature is the most significant predictor ($p < 2e - 16$). For every 1řC increase in water temperature, Dissolved Oxygen (DO) drops by approximately 0.27 mg/L. This is the physical baseline you've correctly controlled for.

The Nitrogen "Penalty": Total Kjeldahl Nitrogen has a significant negative effect (-1.31, $p = 0.011$). In the "Before" (baseline) phase, increasing nitrogen significantly reduces available oxygen.

The "Storm Phase After" Impact: Interestingly, Storm_PhaseAfter is highly significant and negative (-2.00, $p = 3.47e - 06$). This suggests that after the storm passed, DO levels were 2 mg/L lower than the baseline even after accounting for temperature and nutrients. This points to a "lingering hangover" effect—potentially from decaying organic matter settled on the riverbeds.

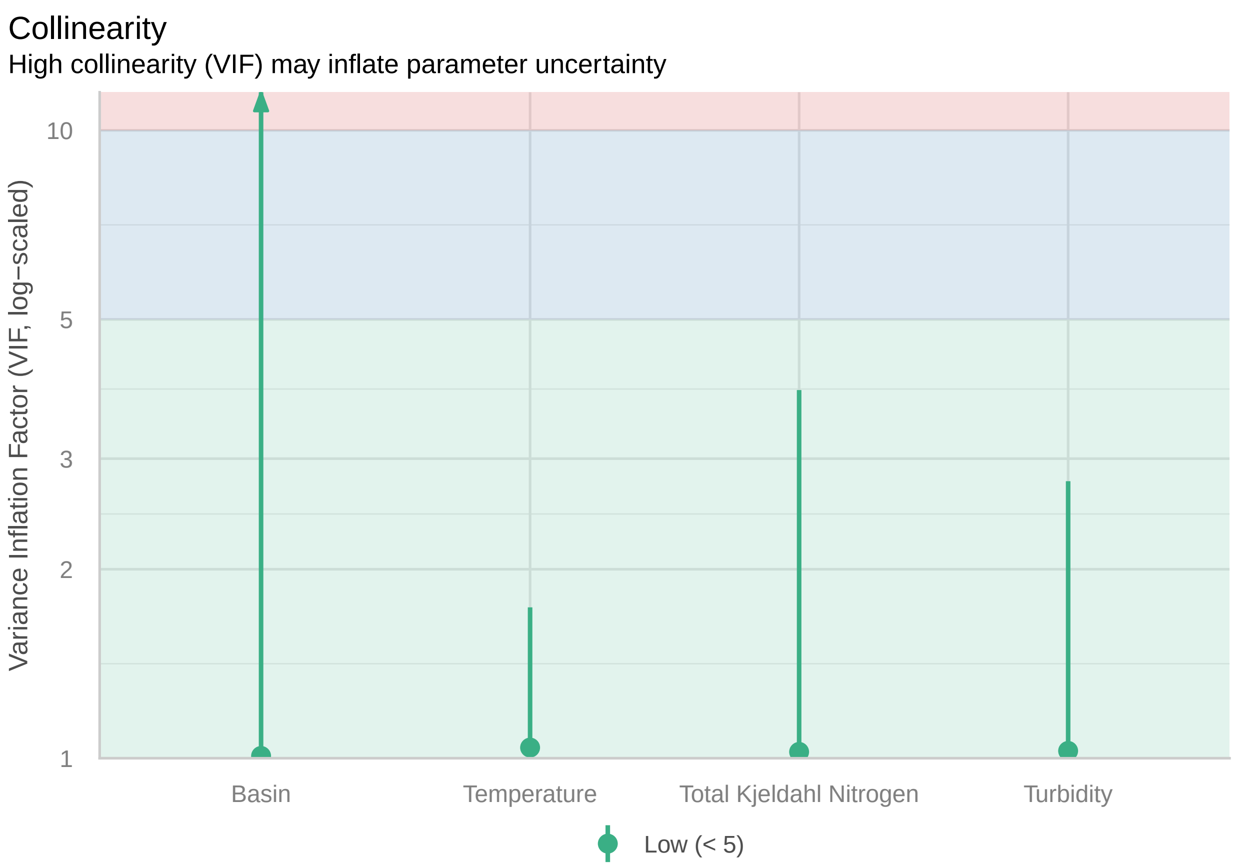
The Interaction Term (TKN:Storm_PhaseAfter): This is significant and positive (1.33, $p = 0.042$). It suggests that in the "After" phase, the negative impact of TKN on DO was actually mitigated compared to the "Before" phase. This could be because the storm "flushed" the system so thoroughly that the remaining nitrogen wasn't as biologically active, or the relationship was drowned out by the overall drop in DO.

Validating model

```
# Calculate VIF
model_vif <- check_collinearity(slope_model_tn)

# Print the results
print(model_vif)
```

```
## # Check for Multicollinearity
##
## Low Correlation
##
##                      Term  VIF      VIF 95% CI adj. VIF Tolerance
##                     Basin 1.01 [1.00, 3670.17]     1.00      0.99
##               Temperature 1.04 [1.00,    1.74]     1.02      0.96
##                 Turbidity 1.03 [1.00,    2.76]     1.01      0.97
##   Total Kjeldahl Nitrogen 1.02 [1.00,    3.86]     1.01      0.98
##   Tolerance 95% CI
##       [0.00, 1.00]
##       [0.58, 1.00]
##       [0.36, 1.00]
##       [0.26, 1.00]
```

```
# Plot the VIF
plot(model_vif)
```



All the VIFs are near one, which indicates that multicollinearity is unlikely for our model.